

The Executor Robot

The Living Map · TSY14 Technical Challenge · Design brief and specification

Written on **04/10/2026** · Official Phase 1 submission deadline: **05/10/2026** · Companion to the “Phase 1 Project Status & Technical Handover” and the “Outside Network Area” brief

Who this is for

- The person who owns the Executor robot in the team (the AI assumed one of four roles: Writer, Beacon, Outside Network, **Executor**) — and any new AI agent helping them.
- It explains what the Executor must do, what the organizers require, what has been designed so far, and what is still open.
- **Nothing for the Executor exists yet: no code, no simulation, no hardware chosen.** The only related code is the Writer simulation, which produces the beacons the Executor will read.
- Labels: **[CONFIRMED]** official document or user · **[PROPOSED]** suggested by the AI, not approved · **[ASSUMPTION]** invented to make the design work · **[OPEN]** no decision yet · **[DESIGN ONLY]** not implemented · **[INTERPRETATION]** the AI’s reading of the documents.

1. What the Executor is

The Executor is the **second robot**. It enters the dangerous space **after** the Writer, with a mission, and finishes the job **without starting from zero**, because the Writer’s knowledge was left behind in beacons. The Writer is “first inside, alone”; the Executor “enters with a mission” (slide 3). **[CONFIRMED]**

1.1 What the official documents require

Requirement (source)	Meaning
Receives a mission: extinguish · extract · seal · retrieve, tied to the chosen scenario (slide 3)	The mission must make sense for the environment you picked.
Enters informed: no blind search; it already knows what to reach and what to avoid (slide 3)	It must carry knowledge in with it — this is the briefing.
Navigates by beacons: “the beacons are the map”, it listens and follows, no GPS needed (slide 3)	Its navigation must really depend on the beacons the Writer left.
Trusts by age: fresh beacons guide it; stale ones it verifies with its own sensors (slide 3; examples on slide 2)	It must judge each beacon by its age and cross-check with its own sensors.
Use at least 2 physical robots , Writer and Executor (slide 6)	The Executor must be a real robot in the final prototype.
Briefed before entry . Slide 3 says “briefed by the command post”; slide 5 gives the BRIEF role to the Outside Network Area; the rule says no direct robot↔command-post link	[INTERPRETATION] The command post decides the mission; the Outside Network Area prepares and delivers the briefing.
Technical focus: <i>mission briefing · beacon-guided navigation · mission execution</i> (cahier des charges p.2)	These are the three things to show.
Goal: “build an Executor Robot that receives a mission and navigates using inherited beacon information” (p.1)	The word “inherited” is the core idea.

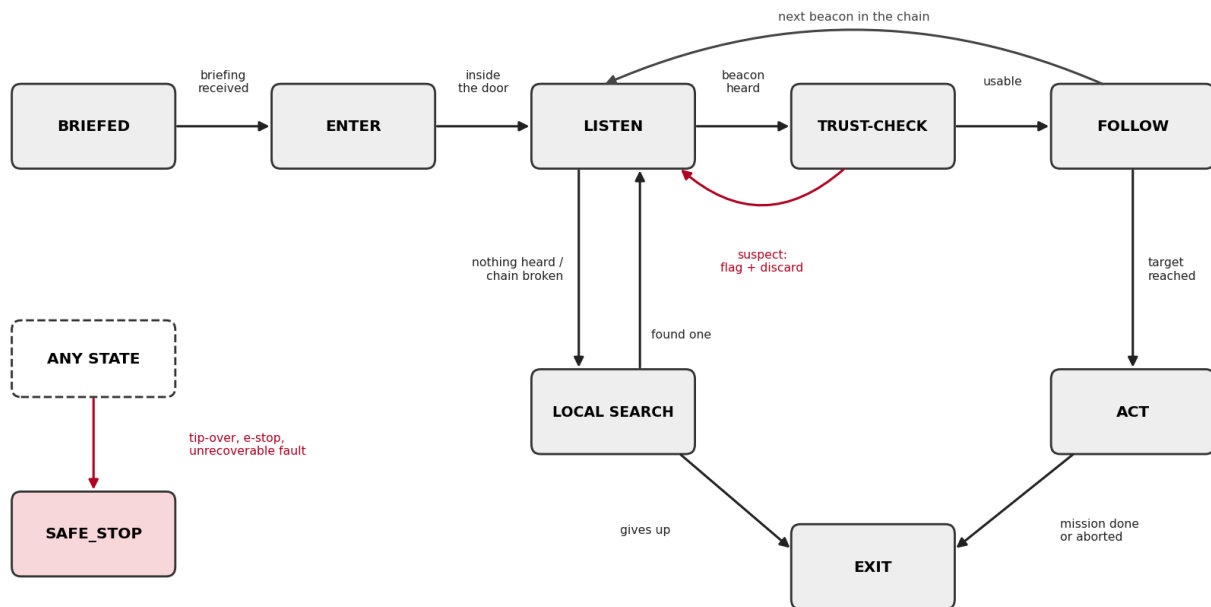
1.2 Points at stake

Phase 1: **Executor robot: 5**, plus shared items it contributes to — technical solution diagrams 5, simulation demo 8, failure cases 3, implementation plan 3. Final phase: it is one of the two robots in the **physical prototype (15 points)**. **Gap to be aware of:** the current simulation covers only the Writer. A demo that never shows the Executor reading beacons would not show the chain the challenge is about.

2. What the Executor depends on

Input / output	From / to	Content	Status
Briefing (I5)	Outside Network Area → Executor, before entry	Mission, target, ordered beacon chain, known hazards, clock synchronisation, trust rules (sketch in the Outside Network brief, section 4.3)	[PROPOSED] [DESIGN ONLY]
Beacon broadcasts (I2)	Beacons → Executor, inside the building	id, type, dir_deg, dist_m, t, ver, and for junctions the 4-bit exits mask (handover PDF, section 8.1)	[PROPOSED] ; radio technology [OPEN]
Own sensors	On the robot	LiDAR, gyro and wheel odometry, a thermal sensor, a floor sensor — to move safely and to cross-check beacons	[PROPOSED] ; nothing chosen
Movement and mission action	Executor → the world	Drives to the target and performs the mission (simulated in Phase 1)	[PROPOSED]
Report back	Executor → anyone	The documents do not require the Executor to send anything out. A decision log (what it trusted, discarded, verified) is useful for judges.	[PROPOSED] optional

3. Behaviour: how the Executor works



EXIT = leave through the entrance by retracing its own path. Whether the Executor must come back out is not specified in the documents (OPEN).

Figure 1 — Proposed Executor state flow (design only). The main path is BRIEFED → ENTER → LISTEN → TRUST-CHECK → FOLLOW → ACT → EXIT, with a loop back to LISTEN for each beacon of the chain.

State	What happens (proposed)	Leaves when
BRIEFED	Waits at the entrance. Receives the briefing and the clock synchronisation, checks the briefing is complete, acknowledges.	Briefing valid. If none arrives it does not enter.
ENTER	Starts its private frame at the entrance: position (0, 0), heading = direction of entry, the same convention as the Writer. Calibrates the gyro while standing still, then drives in.	Through the door.
LISTEN	Moves towards the next expected beacon while listening on the radio, avoiding obstacles and holes.	A beacon is heard; or nothing is heard for too long.
TRUST-CHECK	For each beacon: computes its age (now – t), checks its version, compares it with its own sensor, and classes it fresh / aging / stale / suspect (section 4).	Decision made.
FOLLOW	Drives the next leg of the chain (fresh: normal speed; aging: slow and verify; stale: own senses decide).	Next beacon reached, or target reached.
LOCAL SEARCH	Chain broken or beacon missing: explores locally (e.g. wall following as the Writer does) to find a beacon.	Beacon found, or search limit reached.

State	What happens (proposed)	Leaves when
ACT	At the target, performs the mission action. In Phase 1 the action is simulated.	Action done or aborted.
EXIT	Leaves by retracing its own path. Whether the Executor must come back out is not stated in the documents.	[OPEN]
SAFE_STOP	Stops all motion on tip-over, e-stop or an unrecoverable fault.	Manual reset.

4. Trust by age

The slides stress this behaviour strongly (**[INTERPRETATION]** it is probably what the organizers most want to see in the Executor). The slides show the *same beacon* at different ages: **fresh (2 min)** “trust as measured”; **aging (15 min)** “possible hazard — verify before crossing”; **stale (40 min)** “old warning — rely on your own senses”; **suspect (contradicted)** “live sensor disagrees — flag it, discard”. **[CONFIRMED]**

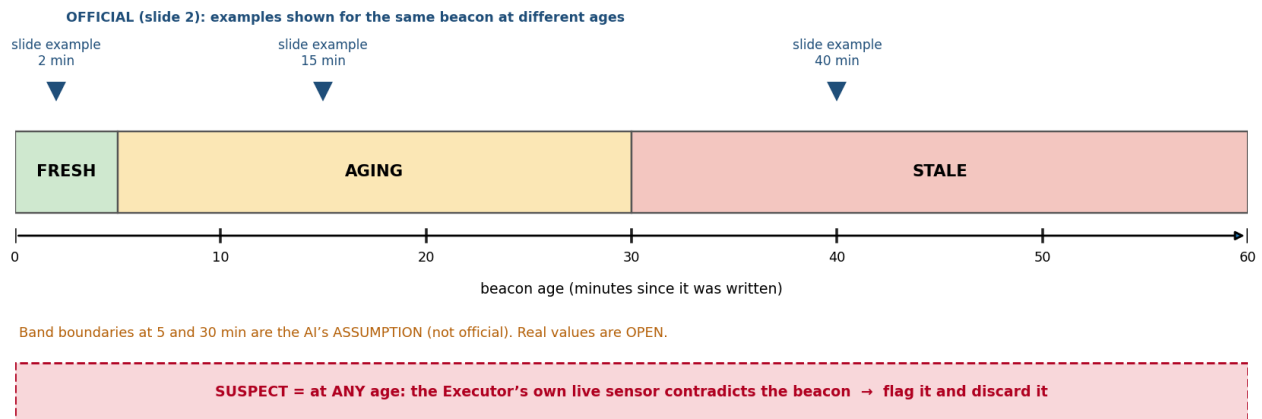


Figure 2 — Trust states over time. The 2 / 15 / 40 min markers are the slide's examples, not official thresholds. The band boundaries (5 and 30 min) are an AI assumption.

4.1 Rules **[PROPOSED]**

State	How it is decided	Executor behaviour
Fresh	Age below the fresh limit; no contradiction	Follow the beacon as measured, at normal speed.
Aging	Age between the fresh and stale limits	Slow down; verify with its own sensor <i>before crossing</i> the area the beacon describes.
Stale	Age above the stale limit	Treat the beacon as a hint only; its own sensors decide what to do.
Suspect	Its own live sensor contradicts the beacon, at any age	Flag it, discard it, record the reason.

4.2 What “contradicted” means for each event type **[PROPOSED]**

Beacon type	Own sensor used	Contradiction example
HAZARD (heat)	Thermal sensor	The beacon says a heat source is ahead, but the Executor, near that spot, reads only ambient temperature. The source may have gone out, or the position may be wrong. The reverse (heat where no beacon warned) is a new hazard to log.
HOLE (missing floor)	Floor-facing distance sensor	The beacon says no floor ahead, but the floor is there (or the reverse). Keep the floor sensor on at all times , even for trusted beacons: it is a safety sensor.
JUNCTION	LiDAR	The beacon's exits mask lists open directions that the LiDAR does not see.
WAYPOINT	—	No event to contradict; only age and position matter.

4.3 Points still open about aging

- **Thresholds [OPEN]**: the slides give examples at 2 / 15 / 40 min; the AI suggested 5 and 30 min. Nothing is coded. The AI's reasoning: they should depend on how fast the scenario changes (a fire changes faster than a mine).
- **Who ages the message? [OPEN]** The cahier des charges lists a “message aging mechanism” under *Beacon Design*, which can mean the beacon itself degrades its message over time. The current proposal is the opposite: the Executor computes the age from the timestamp. This needs agreement with the Beacon teammate.

- **Version and overwriting.** The slide says the version “lets a later agent overwrite” the beacon. **[INTERPRETATION]** This could mean the Executor should be able to *re-write* beacons with updated information (for example “hazard extinguished”). That would require a radio transmitter on the Executor. **[OPEN]** — it is not required by the rules, but it would be a strong demonstration of the idea.
- **Clocks.** Age only works if the Writer’s and Executor’s clocks agree. **[ASSUMPTION]** Both are synchronised to GPS time before entry (see the Outside Network brief).

5. Navigating by beacons — the hard part

5.1 What a beacon actually tells the Executor

The current beacon message (handover PDF, section 8) carries `dir_deg` and `dist_m`, but their meaning depends on the type: for HAZARD, the bearing and a *rough* range to the heat source; for HOLE, the direction of the missing floor and 0.3 m; for JUNCTION and WAYPOINT, the Writer’s heading at the drop and a **placeholder** distance (3.5 m) that is **not** the real distance to the next beacon. The reason: when the Writer drops a beacon, it does not yet know where the next one will be.

[OPEN DESIGN ISSUE] Beacons only, or beacons plus briefing? Because of the point above, the true geometry between beacons (where the next beacon is, from this one) can only be computed from the Writer’s complete log, by the Outside Network, and delivered in the briefing. So in the current design the Executor depends on **both** the briefing (geometry) and the beacons (confirmation, age, trust). A judge may ask: “is the memory really in the space?” The team must decide whether this is acceptable, or whether the beacons should carry more (for example the direction and distance *back* to the previous beacon, which the Writer does know at drop time).

5.2 Proposed approach **[PROPOSED]** **[DESIGN ONLY]**

- The briefing gives the ordered chain, with each beacon’s position in the shared private frame and the expected direction and distance to the next one.
- The Executor drives towards the next expected beacon using dead reckoning (wheels and gyro) with obstacle avoidance, and starts listening.
- When a beacon is heard it is judged (section 4). A trusted beacon can also **correct the robot’s drift**: its position is known from the briefing, so the Executor can reset its estimated position when it is at the beacon.
- The same heading correction as the Writer (using straight walls) can be reused. It has the same weakness: it assumes the walls run along the x and y axes.

5.3 How does it know it is at a beacon? **[OPEN]**

Option	Idea	Trade-off
A. “Heard” = in range	A beacon is “reached” when it is heard at all	Simplest; the precision equals the radio range, which is probably several metres.
B. Signal strength	Use how strong the signal is: stronger means closer; follow the increase	Cheap, but indoor signal strength is noisy; must be measured on hardware.
C. A ranging radio (e.g. ultra-wideband)	Measures distance directly	Most precise; more and costlier hardware on both beacons and robot.

None of these has been tested. How radio behaves inside a building is unknown and **must be measured** before the design is trusted. The choice depends on the radio technology the Beacon teammate selects.

5.4 If there is no briefing chain (fallback)

If the Writer is lost, no log exists and the briefing contains only the mission. The Executor could still follow each heard beacon’s direction, and use the junction exits mask to choose branches. This works only in simple layouts: at a junction the direction stored in the beacon is the way the Writer happened to continue, which may not lead to the target. **[ASSUMPTION]** This is a weak mode and should be shown as such.

6. Hardware (everything proposed, nothing chosen)

Function	Need	Note
Drive base and odometry	Two driven wheels with encoders and an IMU for heading	Could copy the Writer's design so that code can be shared.
Main computer	Runs the trust logic, the chain follower and the sensors	The UNO Q (as on the Writer) would allow reuse of the same software structure. [OPEN]
Obstacle and junction sensing	LiDAR	Same role as on the Writer; also used to check the junction exits.
Heat verification	Thermal sensor	Needed to contradict or confirm a HAZARD beacon. Not chosen.
Floor verification and safety	Floor-facing distance sensor	Needed to confirm a HOLE beacon and to avoid falling.
Beacon receiver	A radio that matches the beacons	Depends on the Beacon teammate's choice. [OPEN]
Beacon transmitter (optional)	To re-write beacons (version overwrite)	Only if the team decides the Executor updates beacons.
Link for the briefing	To receive the briefing at the entrance	Depends on the Outside Network design. [OPEN]
Mission tool	The device that "extinguishes", "seals", "extracts" or "retrieves"	Not chosen. For a demo, a safe stand-in is advised (an indicator light, a servo arm or a flag). Real fire or water near the electronics is not recommended.
Power and safety	Battery, a separate rail for actuators, an emergency stop	As for the Writer.

6.1 What can be reused from the Writer code

Reusable	New for the Executor
The hardware-abstraction structure · wall following and obstacle avoidance · gyro calibration and the wall-based heading correction · stuck detection · tip-over and battery safety · logging of decisions · retracing its own path	A way to listen to beacons (the Writer's interface has no "listen" function; one would have to be added) · the trust module · the chain follower · the briefing reader · the mission action · optional beacon re-writing

This is a suggestion; the code has not been adapted. **[PROPOSED]**

7. Phase 1 simulation plan **[NOT STARTED]**

- Use the **same simulated building** and the beacons left by a Writer run.
- Reception model: a beacon is heard when the robot is within a chosen radius. **[ASSUMPTION]** — no radio physics is simulated.
- The file `beacons_on_ground.json` contains `true_x` and `true_y`, which exist only for plotting. **The Executor must not use them:** they are the simulation's hidden ground truth. It should use only what would be on the air, plus the positions given in the briefing.
- Time can be shifted in the simulation to test aging.
- The Writer simulation is not fully reliable yet. Seeds without a flagged problem in the earlier tests were 2, 4, 5, 6, 8, 11, 12 and 13, or a hand-made beacon set can be used.
- Suggested success criterion: the Executor reaches within 0.5 m of the target and performs the action, never enters a hole, and logs every trust decision. **[ASSUMPTION]**

Test	Setup	Expected result	Status
E1 Normal	Fresh beacons, valid briefing	Follows the chain, reaches the target, acts	[NOT STARTED]
E2 Aging / stale	Clock moved forward so beacons become aging, then stale	Aging: slows and verifies. Stale: relies on its own sensors. Still reaches the target	[NOT STARTED]
E3 Contradicted	The heat source is removed after the Writer's run	Thermal reading is ambient: the HAZARD beacon is flagged suspect and discarded; the outcome is reported	[NOT STARTED]
E4 Missing beacon	One beacon of the chain removed	Falls back to local search; continues or gives up cleanly	[NOT STARTED]
E5 No log	Briefing contains the mission only	Weak beacon-only mode: reaches the target or stops safely with a clear status	[NOT STARTED]

Test	Setup	Expected result	Status
E6 Heading drift	Larger gyro bias	Heading correction and beacon position resets keep it on course; shows the limit	[NOT STARTED]
E7 Version	Two versions of the same beacon	The higher version wins	[NOT STARTED]

8. Failure cases [DESIGN ONLY]

What fails	Consequence	Planned response
A beacon is stale	Old information may be wrong	Use as a hint only; own sensors decide
A beacon is contradicted by a sensor	Wrong guidance or a false alarm	Flag, discard, record; continue with the next beacon
A beacon in the chain is missing or dead	The chain is broken	Fall back to the last direction and distance, then local search; give up after a limit
No beacon can be received (radio failure)	The Executor is blind	Rely on dead reckoning and the briefing; stop safely if lost
Heading drifts or is lost	The robot steers off course	Wall-based correction and beacon position resets; stop if the estimate is clearly wrong
Stuck on an obstacle the LiDAR cannot see	No progress	Back up and turn; after repeated stalls, abort and exit
The target is blocked (for example by a hole)	The mission cannot be completed	Report “target unreachable”; do not cross
Briefing not delivered or corrupted	Entering without a plan	Do not enter until a valid briefing is acknowledged
Clocks not synchronised	Ages are wrong, so trust is wrong	Synchronise from GPS time before entry; log the sync
Low battery	Mission cannot finish	Abort and exit while energy remains
Tip-over	Danger to the robot	SAFE_STOP

9. Open decisions

“Responsible” assumes the four-role split; real names are unknown. No decision below has been made.

Item	Current status	Decision needed	Responsible
Event types the Executor must handle	HAZARD, JUNCTION, HOLE coded on the Writer	Confirm; each one needs a sensor on the Executor to cross-check it	Writer owner + Executor owner
Aging thresholds	Slide examples 2 / 15 / 40 min; AI proposal 5 / 30 min	Choose values; depend on the scenario	Beacon + Executor owners
Who ages the message	Reader-side in the current proposal; the cahier lists a “message aging mechanism” under Beacon Design	Decide: beacon degrades its message, or the Executor computes the age	Beacon + Executor owners
Beacons only vs briefing-assisted	Briefing-assisted in the current design	Accept, or put more geometry on the beacons	Whole team
Proximity method and receiver radio	Three options in 5.3; none tested	Choose after radio tests	Beacon + Executor owners
Heading consistency	Shared private-frame convention; method not decided	Wall correction, beacon resets, or both	Executor + Outside Network owners
Does the Executor re-write beacons	Not required; the slide’s word “overwrite” suggests it	Decide; needs a transmitter and a write protocol	Beacon + Executor owners
Mission and physical stand-in	Demo mission “reach the hot spot and extinguish”, simulated	Confirm the mission and a safe tool	Whole team
Must the Executor exit?	Not stated in the documents	Decide; matters for “extract” and “retrieve”	Whole team
Briefing delivery	Not decided	Link and acknowledgement method	Outside Network + Executor owners
Computer and hardware reuse	UNO Q suggested for reuse	Choose	Executor owner

10. What the Executor must deliver, and next steps

	Task	Why
☐	Agree the briefing (I5) with the Outside Network owner, and the beacon message and aging rules with the Beacon owner	Everything else depends on these interfaces
☐	Decide the aging thresholds, who computes the age, and beacons-only vs briefing-assisted	Core of “trust by age”
☐	Write a simulated Executor (reception, trust module, chain follower, mission action) and run tests E1-E7	Executor robot (5) and the simulation demo (8)
☐	Draw the Executor architecture and data-flow diagrams (state flow in Figure 1 is a draft)	Technical solution diagrams (5)
☐	Write the Executor failure cases (section 8) and record the results of the tests	Failure cases (3)
☐	List the final-phase hardware, a timeline to 01/12/2026, and the risks	Implementation plan (3); physical prototype (15) later
☐	Write the “Outside Network and Executor” part of the 6-page report (about 0.7 pages shared with the Outside Network owner)	Technical report (2)

Not yet done by anyone: opening the two links inside the official cahier des charges — the “System Architecture Diagram” resource (a Google Drive file) and the submission form. The architecture diagram may show what the organizers expect for the Executor. The links are listed in section 2.2 of the handover PDF.